

```
fprintf(&#39;Skip Distance = %.2f km\n&#39;;Skip_Distance);
```

```
fprintf(&#39;Maximum Usable Frequency (MUF) = %.2f MHz\n&#39;,MUF);
```

```
%% Plot Results
```

```
figure(&#39;Name&#39;,...
```

```
&#39;Ground Wave and Sky Wave Propagation&#39;,...
```

```
&#39;NumberTitle&#39;,&#39;off&#39;);
```

```
subplot(2,2,1)
```

```
plot(d,E_ground,&#39;LineWidth&#39;,2)
```

```
grid on
```

```
xlabel(&#39;Distance (km)&#39;)
```

```
ylabel(&#39;Field Strength (mV/m)&#39;)
```

```
title(&#39;Ground Wave Propagation&#39;)
```

```
subplot(2,2,2)
```

```
bar(MUF)
```

```
grid on
```

```
title(&#39;Maximum Usable Frequency&#39;)
```

```
ylabel(&#39;Frequency (MHz)&#39;)
```

```
set(gca,&#39;XTickLabel&#39;,{&#39;MUF&#39;})
```

```
subplot(2,2,3)
```

```
bar(Skip_Distance)
```

```
grid on
```

```
title(&#39;Skip Distance&#39;)
```

```
ylabel(&#39;Distance (km)&#39;)
```

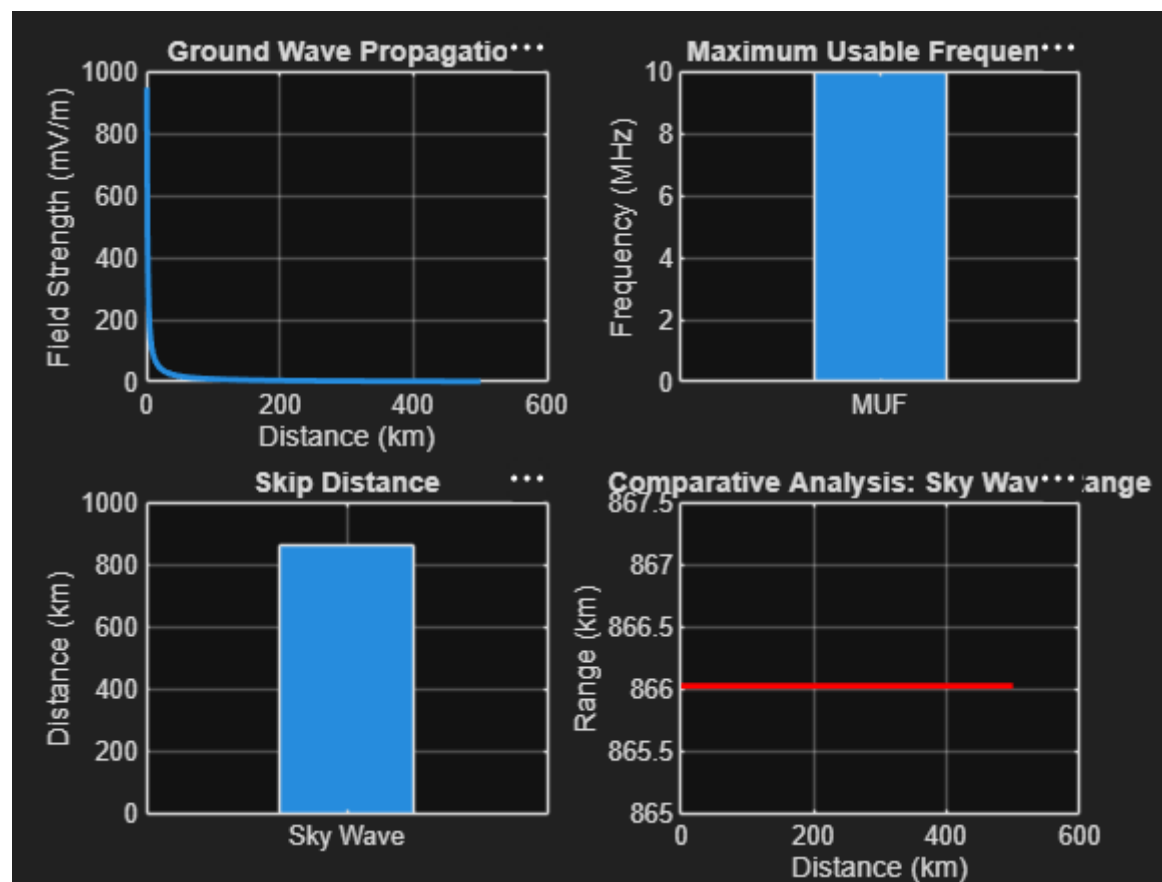
```
set(gca,&#39;XTickLabel&#39;,{&#39;Sky Wave&#39;})
```

```
subplot(2,2,4)
```

```
plot(d)
```

OUTPUT:

```
GROUND WAVE AND SKY WAVE ANALYSIS
|-----|
Transmitted Power = 10.00 kW
Ionospheric Height = 250.00 km
Skip Distance = 866.03 km
Maximum Usable Frequency (MUF) = 10.00 MHz
>>
```



EXPT 9b - Simulation of Space Wave Propagation Characteristics using MATLAB

DATE:06/08/2026

Aim: To simulate space wave propagation and analyze its characteristics using MATLAB.

Overall MATLAB code:

```
clc;

clear;

close all;

%% Objective 1 : Space Wave Propagation

f = 900; % Frequency (MHz)

Pt = 30; % Transmit Power (dBm)

d = 1:1:100; % Distance (km)

FSPL = 32.44 + 20*log10(f) + 20*log10(d);

Pr = Pt - FSPL;

%% Objective 2 : LOS and Radio Horizon

ht = 50; % Transmitter Height (m)

hr = 25; % Receiver Height (m)

d_LOS = 4.12*(sqrt(ht)+sqrt(hr));

%% Objective 3 : Link Budget Analysis

Gt = 10; % Tx Gain (dB)

Gr = 8; % Rx Gain (dB)

Pr_link = Pt + Gt + Gr - FSPL;

%% Display Results

fprintf('%39;\nSPACE WAVE PROPAGATION ANALYSIS\n%39;);
```

```

fprintf(&#39;-----\n&#39;);

fprintf(&#39;Operating Frequency = %.0f MHz\n&#39;,f);


fprintf(&#39;LOS Distance = %.2f km\n&#39;,d_LOS);

fprintf(&#39;Received Power at 10 km = %.2f dBm\n&#39;,Pr(10));

%% Plot Results

figure(&#39;Name&#39;,...

&#39;Space Wave Propagation Characteristics&#39;,...

&#39;NumberTitle&#39;,&#39;off&#39;);

subplot(2,2,1)

plot(d,FSPL,&#39;LineWidth&#39;,2)

grid on

xlabel(&#39;Distance (km)&#39;);

ylabel(&#39;FSPL (dB)&#39;);

title(&#39;Free Space Path Loss&#39;);

subplot(2,2,2)

plot(d,Pr,&#39;LineWidth&#39;,2)

grid on

xlabel(&#39;Distance (km)&#39;);

ylabel(&#39;Received Power (dBm)&#39;);

title(&#39;Received Power vs Distance&#39;);

subplot(2,2,3)

bar(d_LOS)

grid on

title(&#39;LOS Communication Range&#39;);

ylabel(&#39;Distance (km)&#39;);

```

```

set(gca,'XTickLabel',{'LOS Range'})

subplot(2,2,4)

plot(d,Pr_link,'LineWidth',2)

grid on

xlabel('Distance (km)')

ylabel('Power (dBm)')

title('Link Budget Analysis')

```

```

SPACE WAVE PROPAGATION ANALYSIS
-----
Operating Frequency = 900 MHz
LOS Distance = 49.73 km
Received Power at 10 km = -81.52 dBm
>>

```

